

Urinary incontinence and urinary tract symptoms

Companion Figures

Surface markings of the kidney

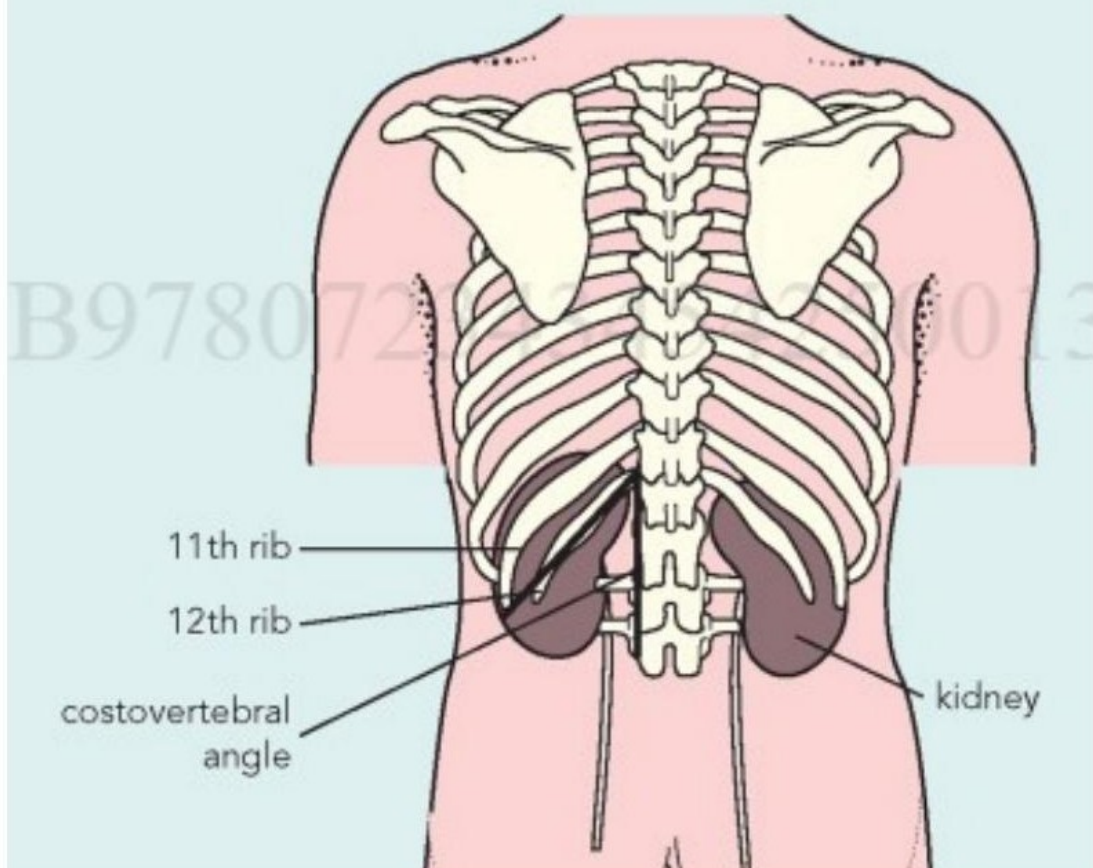
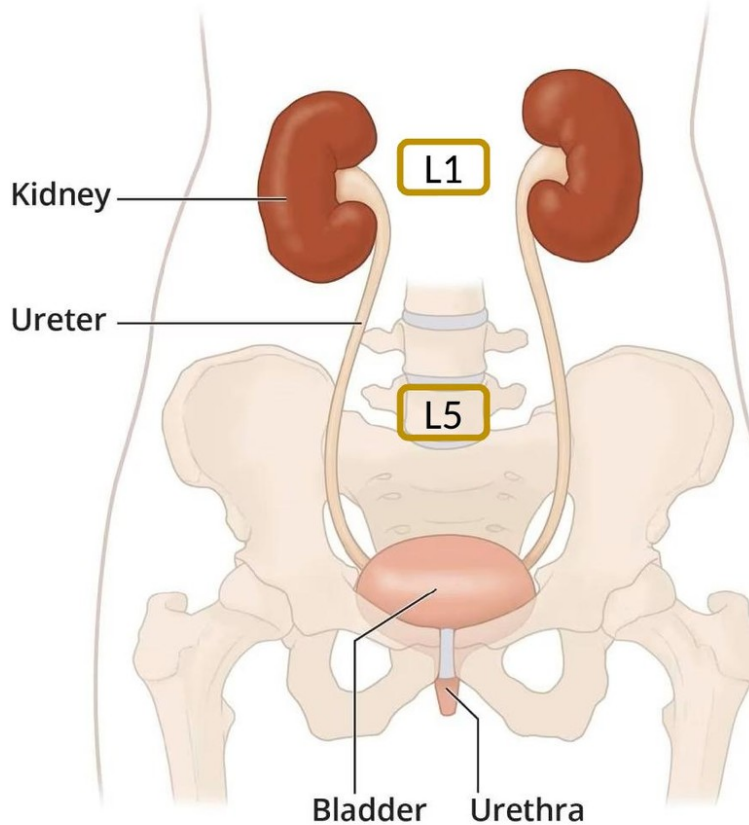
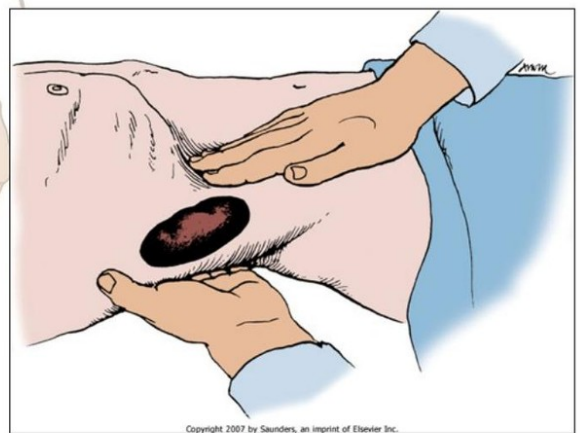


Figure L1.1 — Surface markings of the kidney (11th rib, 12th rib, costovertebral angle)

Urinary Tract



- Kidneys are retroperitoneal.
- Ureters descend in front of the tips of transverse spinous processes, cross sacro-iliac joint, then forwards next to rectum/ vagina.
- Bladder is anterior in pelvis.



structural and cellular organisation of the kidney, bladder, and prostate and relate this to

Figure L1.2 — Urinary tract overview: kidneys at L1, ureters descending to bladder, with palpation/ballotting

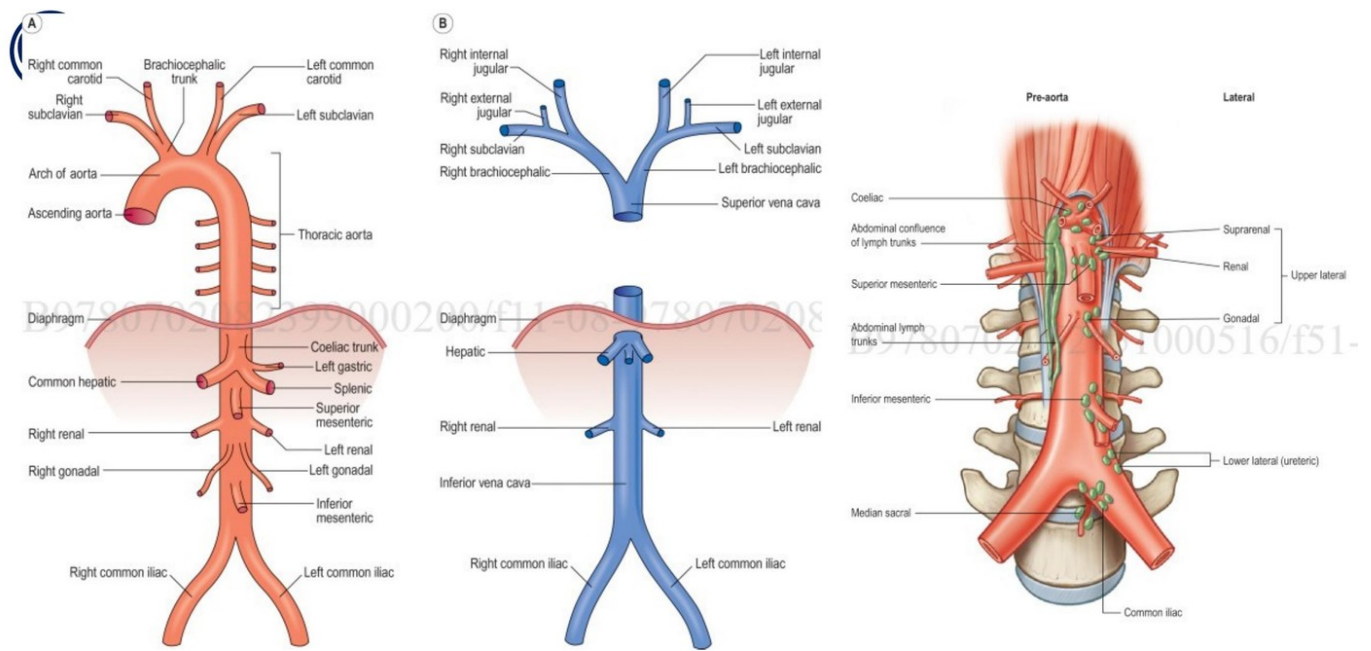


Figure L1.3 — Abdominal aorta, IVC and pre-aortic/lateral abdominal lymph node groups

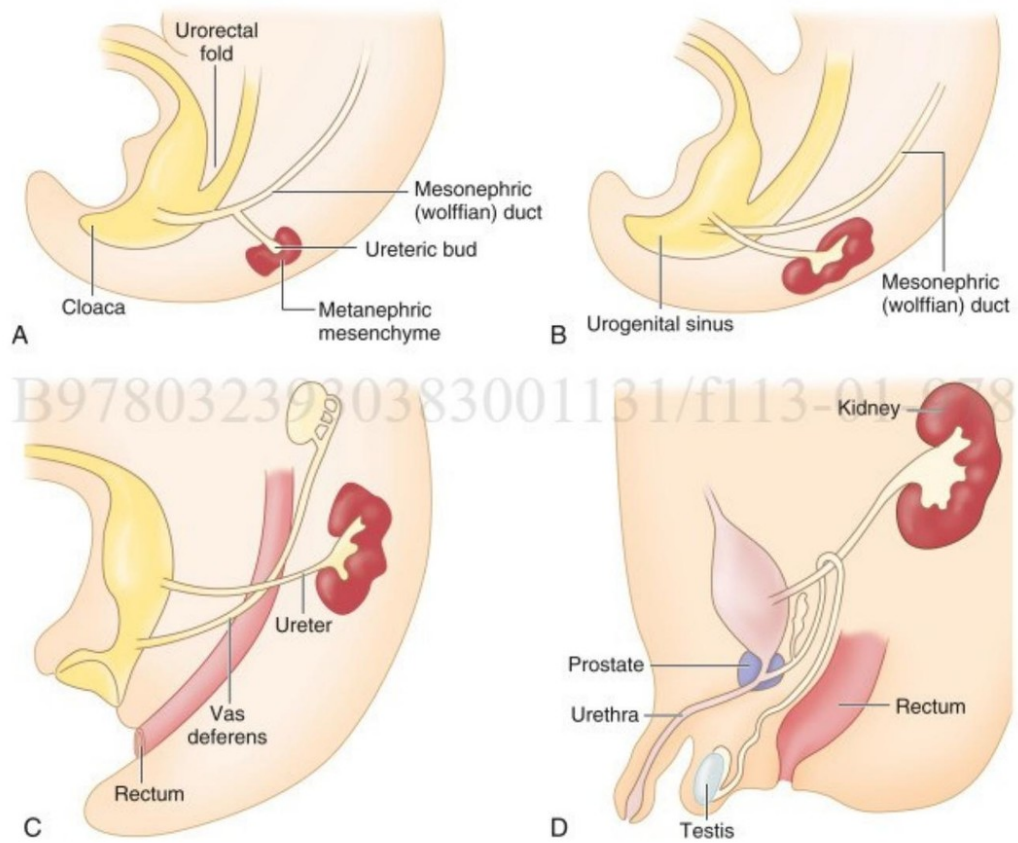
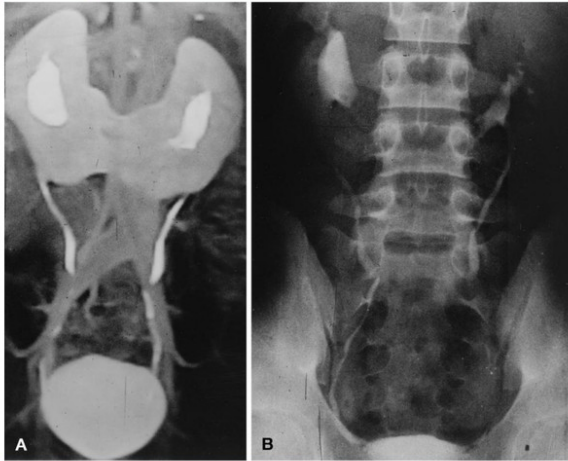


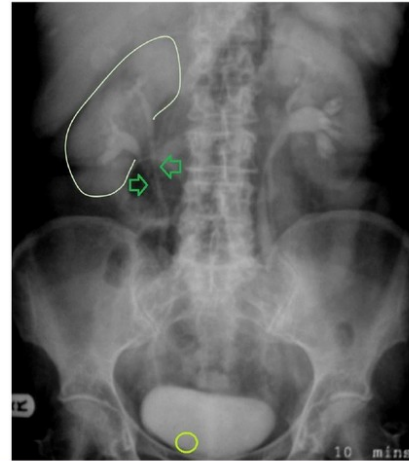
Figure L1.4 — Embryology: cloaca with mesonephric duct + ureteric bud → mature urinary tract

- Horse-shoe kidney
- Ectopic kidney



CT (left) and IVU of horseshoe kidney

- Partial duplication
- Complete duplication



IVU of bilateral partial duplication of ureters

Q-1: Genitourinary structure and function: Describe the structural and cellular organisation of the kidney, bladder, and prostate and relate this to

Figure L1.5 — Anatomic variants: horseshoe kidney (CT + IVU) and bilateral partial ureteric duplication (IVU)



“Hold up” points in the ureter

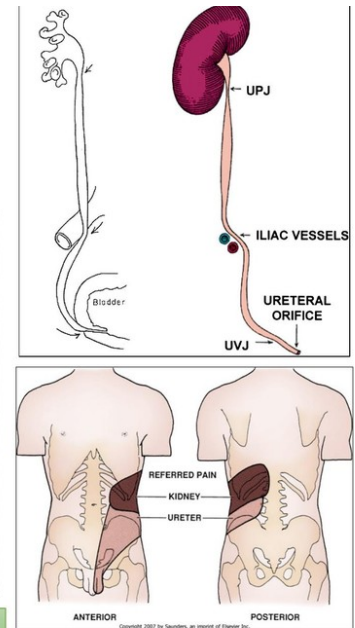
Constriction points may block urine flow, especially if a kidney stone dislodges and becomes a ureteric stone (pain, ipsilateral impaired renal function):

- Where the renal pelvis joins the top of the ureter- pelvic ureteric junction (PUJ, or UPJ)
- Pelvic brim, crossing the iliac vessels
- As it passes through the bladder wall; uretero-vesical junction (UVJ, or VUJ)



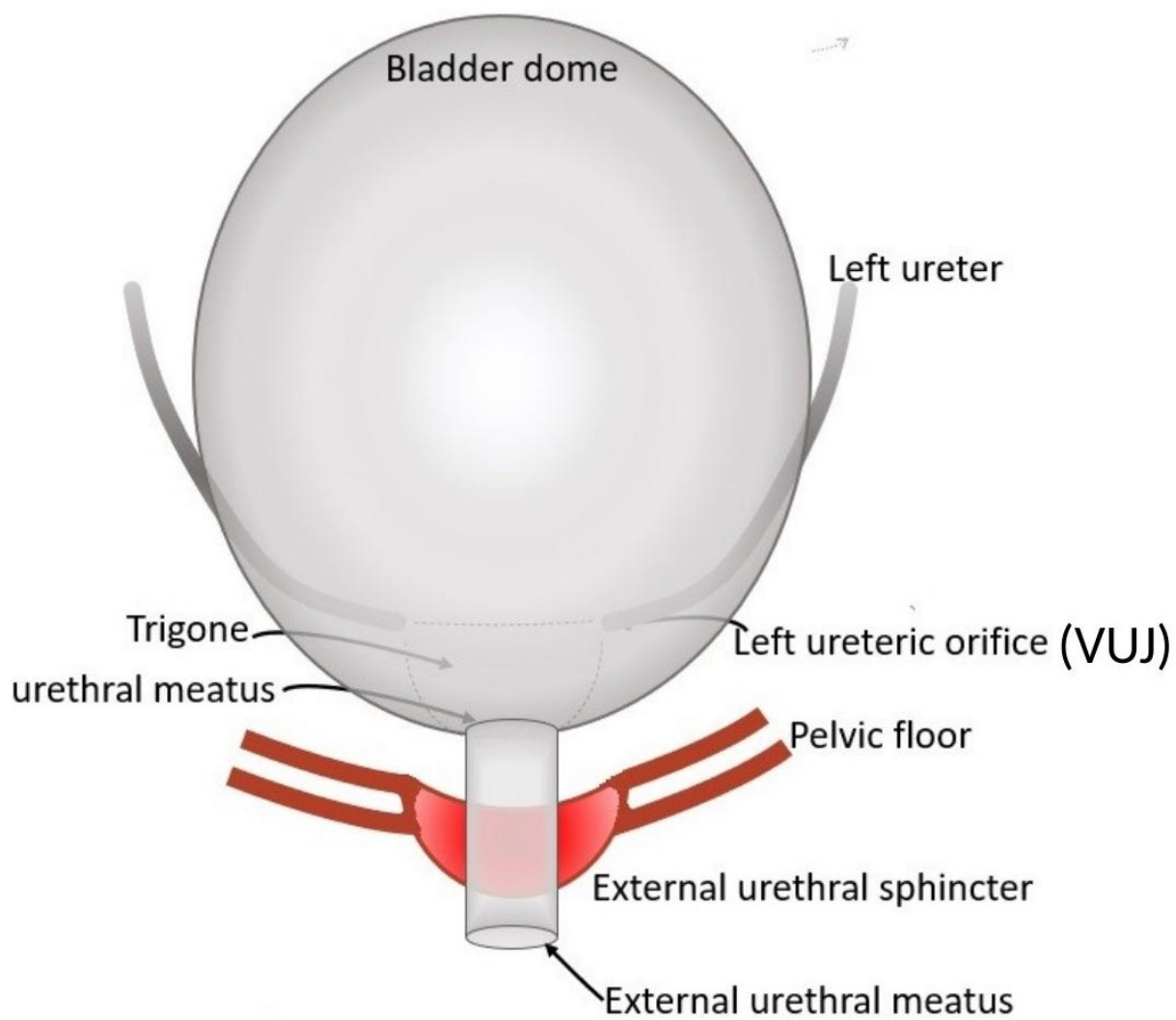
Idiopathic right PUJ obstruction (*delayed nephrogram, swollen*)

Right VUJ obstruction (*ureterocoele*)



1-BRS-URO-3: Genitourinary disorders: Summarise the pathology and pathophysiology of genitourinary disorders.

Figure L1.6 — Three ureteric “hold-up” points (PUJ, pelvic brim, VUJ), illustrative obstruction films and referred pain pattern



ganisation of the kidney, bladder, and prostate and relate this to

Figure L1.7 — Bladder anatomy: dome, trigone, ureteric orifice (VUJ), internal/external urethral meatus, external sphincter

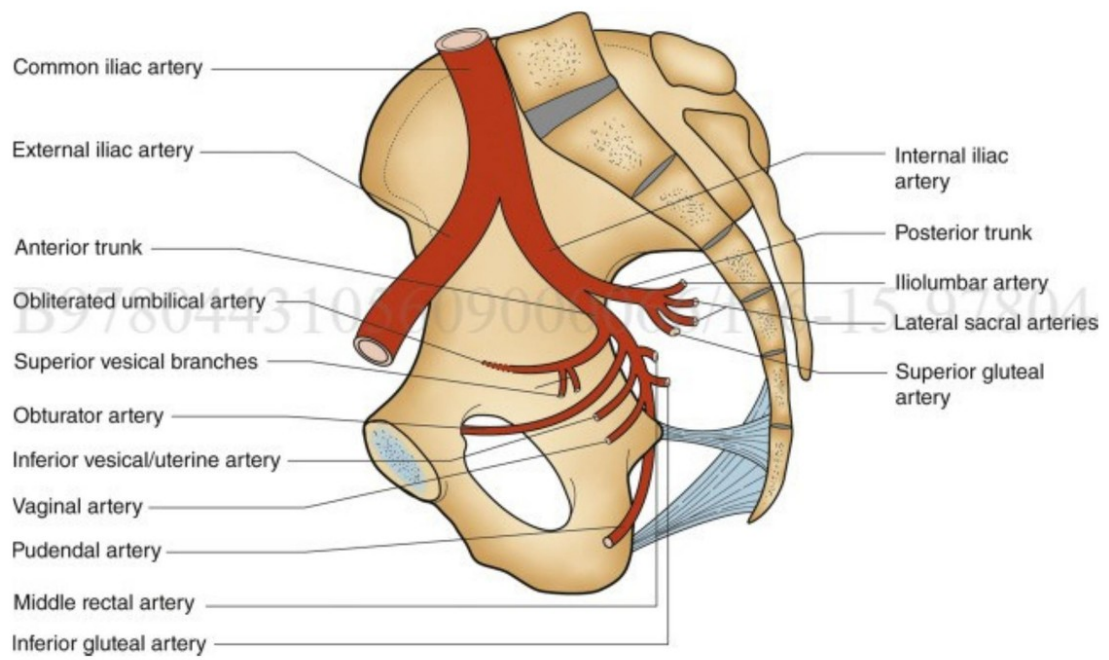


Figure L1.8 — Internal iliac artery: anterior trunk (umbilical, vesical, obturator, vaginal/uterine, pudendal, rectal, gluteal) and posterior trunk (iliolumbar, lateral sacral, superior gluteal)

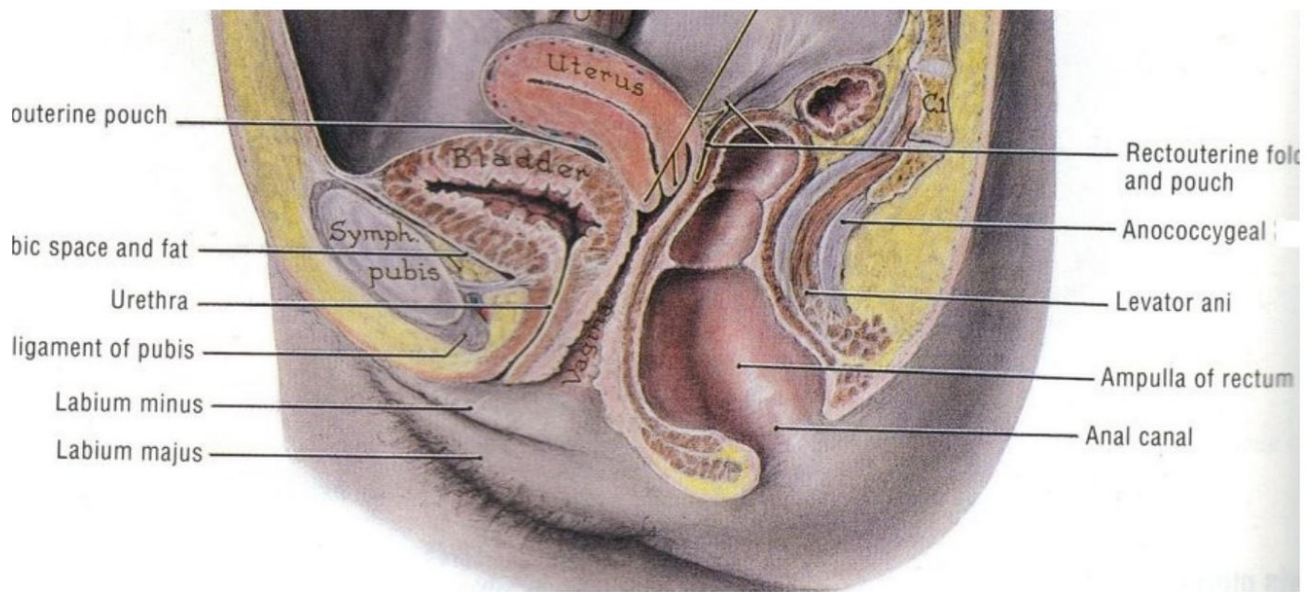
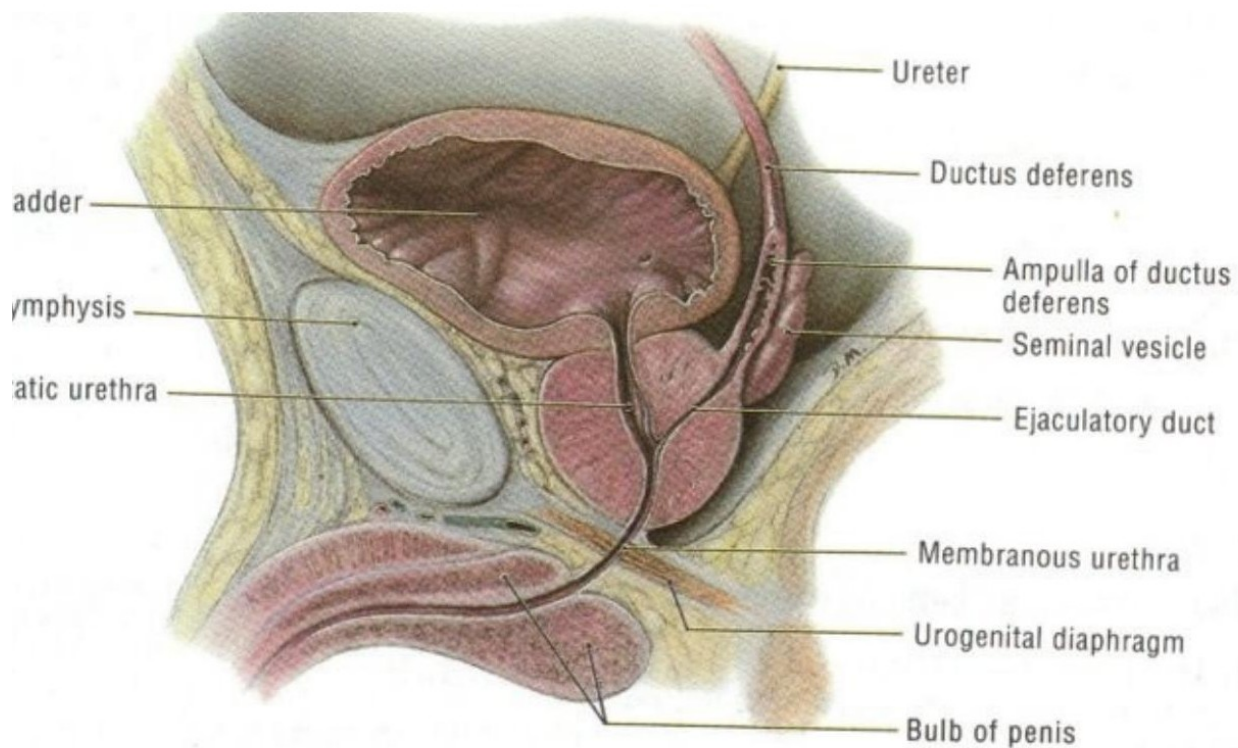


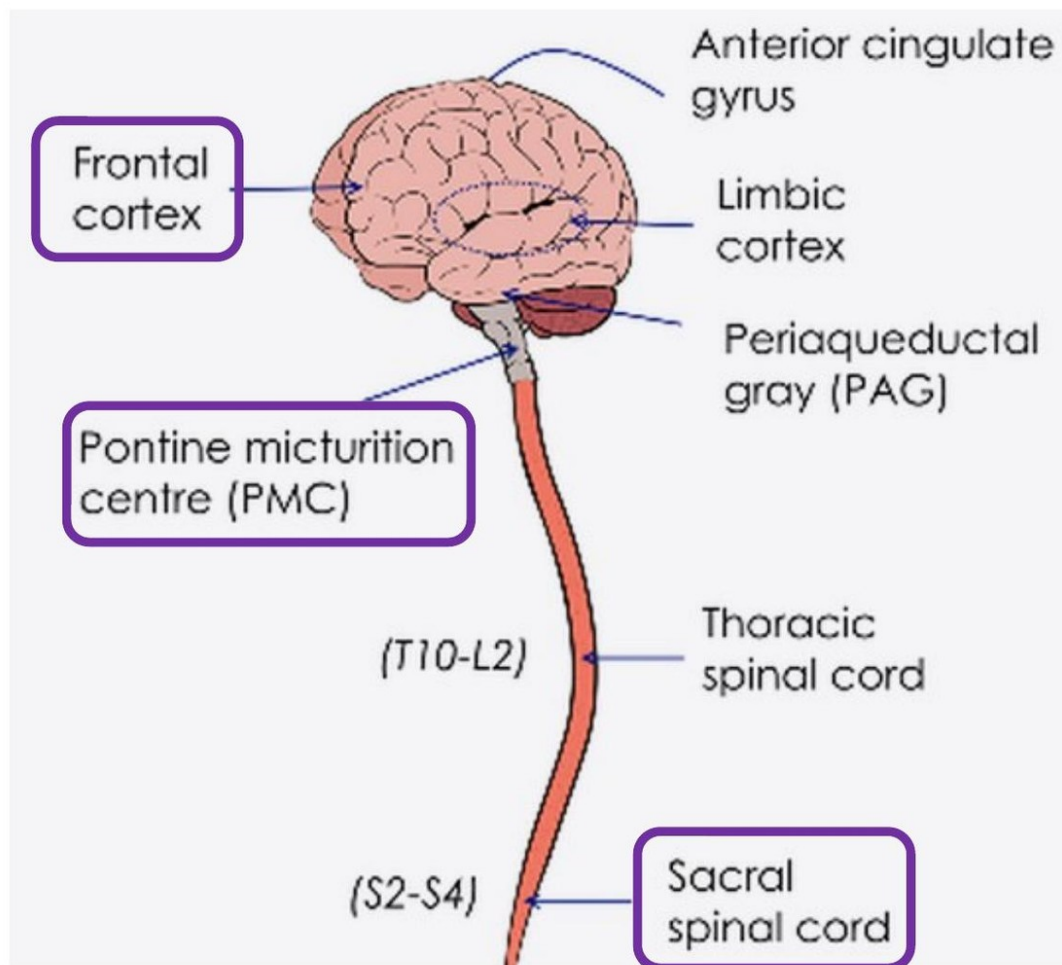
Figure L1.9 — Female pelvis sagittal: bladder, urethra, vesicouterine pouch, retropubic space, levator ani, anal canal



Cellular organisation of the kidney, bladder, and prostate and relate this to

Figure L1.10 — Male pelvis sagittal: bladder, prostate, prostatic/membranous urethra, ductus deferens, ejaculatory duct, seminal vesicle

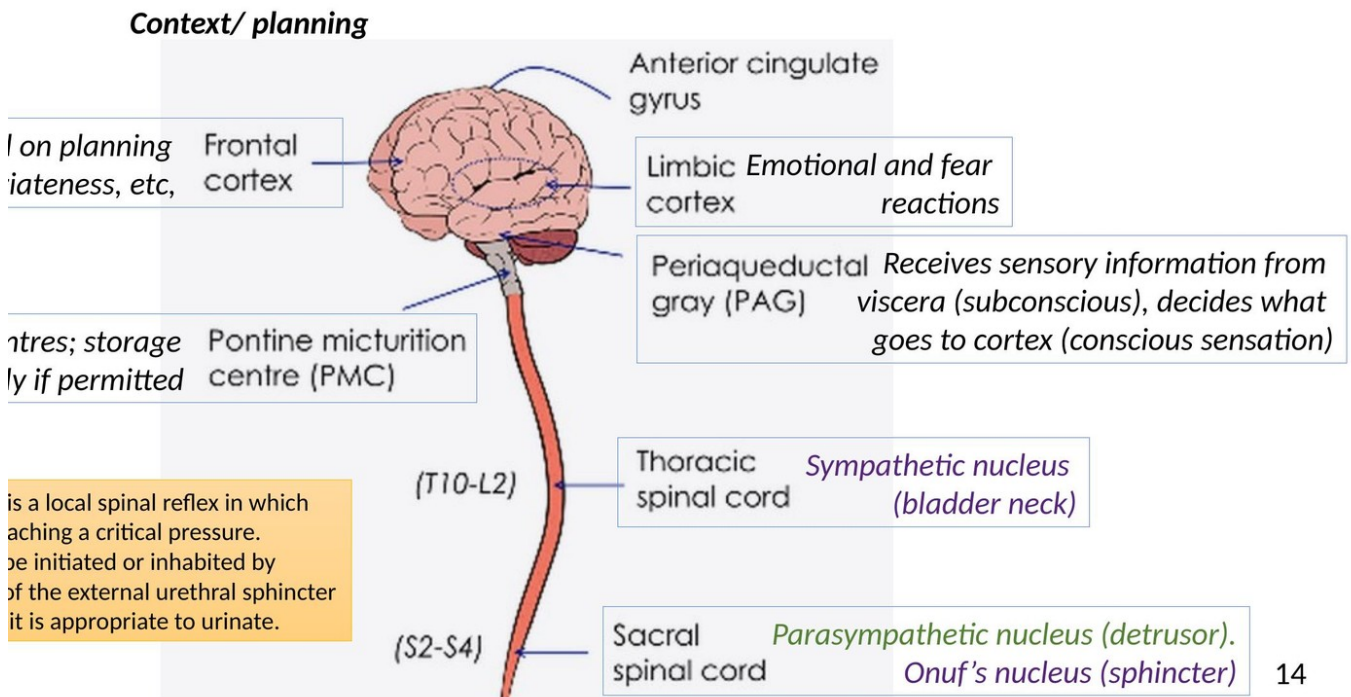
Neurulation in summary



al and cellular organisation of the kidney, bladder, and prostate and relate this to

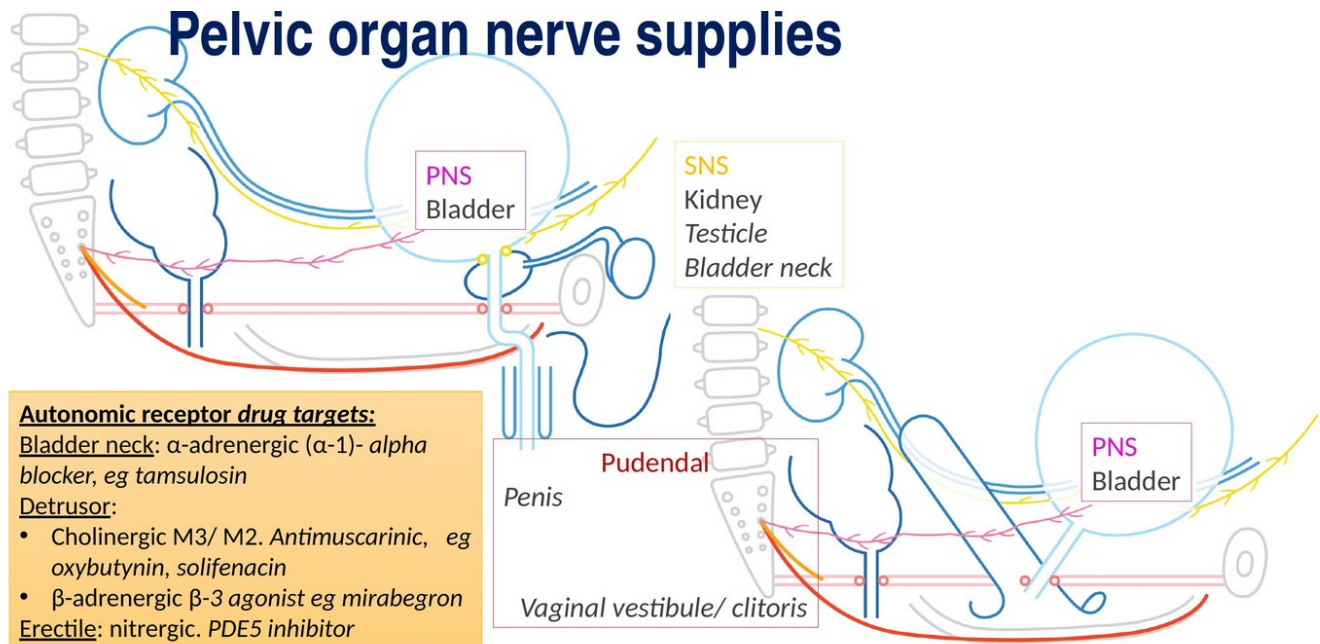
Figure L1.11 — Neural control of micturition: cortex, limbic, PAG, PMC; T10–L2 sympathetic; S2–S4 parasympathetic + Onuf

physiology of micturition



Structure and function: Describe the structural and cellular organisation of the kidney, bladder, and prostate and relate this to

Figure L1.12 — Functional roles: prefrontal (planning), limbic (emotion), PAG (visceral relay), PMC (switch), spinal nuclei



1-BRS-URO-1: Genitourinary structure and function: Describe the structural and cellular organisation of the kidney, bladder, and prostate and relate this to function.

Figure L1.13 — Pelvic organ innervation and autonomic drug targets (α 1, M2/M3, β 3, PDE5)

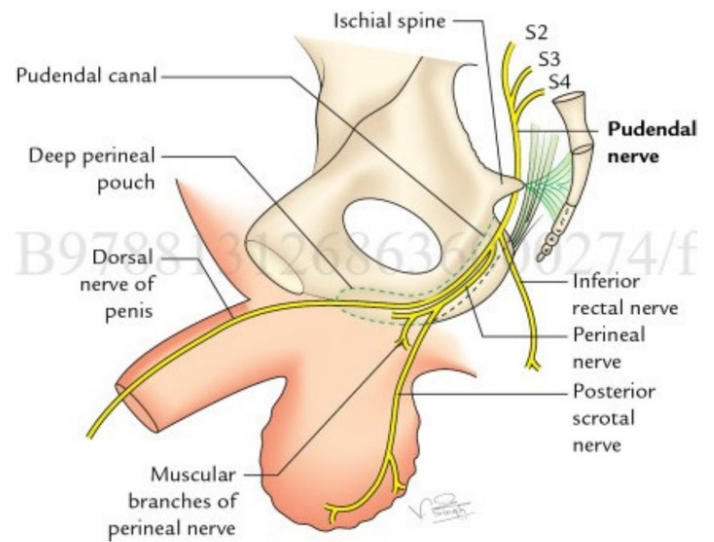
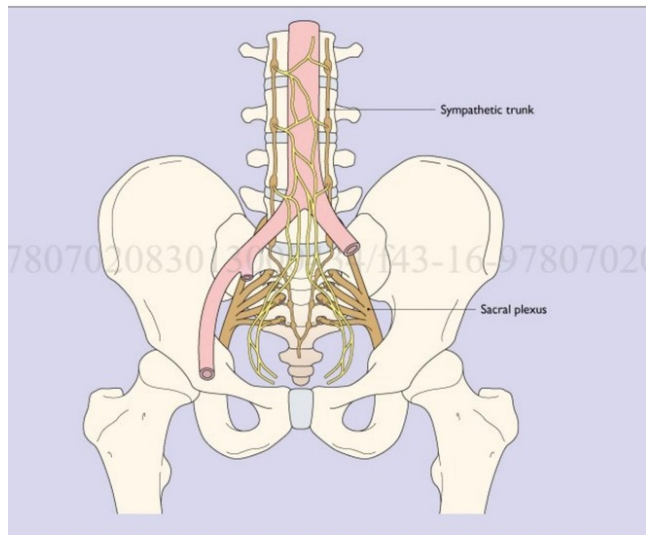


Figure L1.14 — Sacral plexus and pudendal nerve (S2–S4) branches: dorsal nerve of penis, inferior rectal, perineal, posterior scrotal

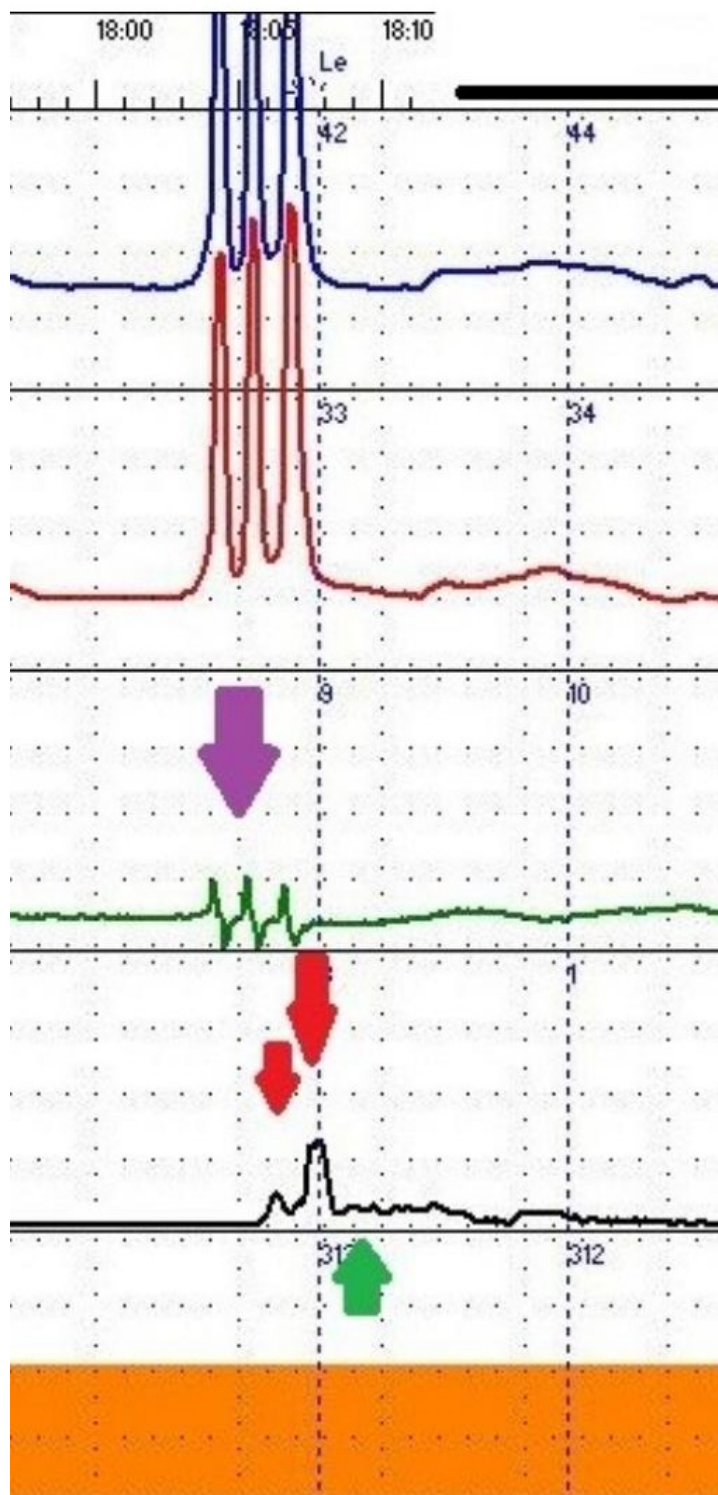
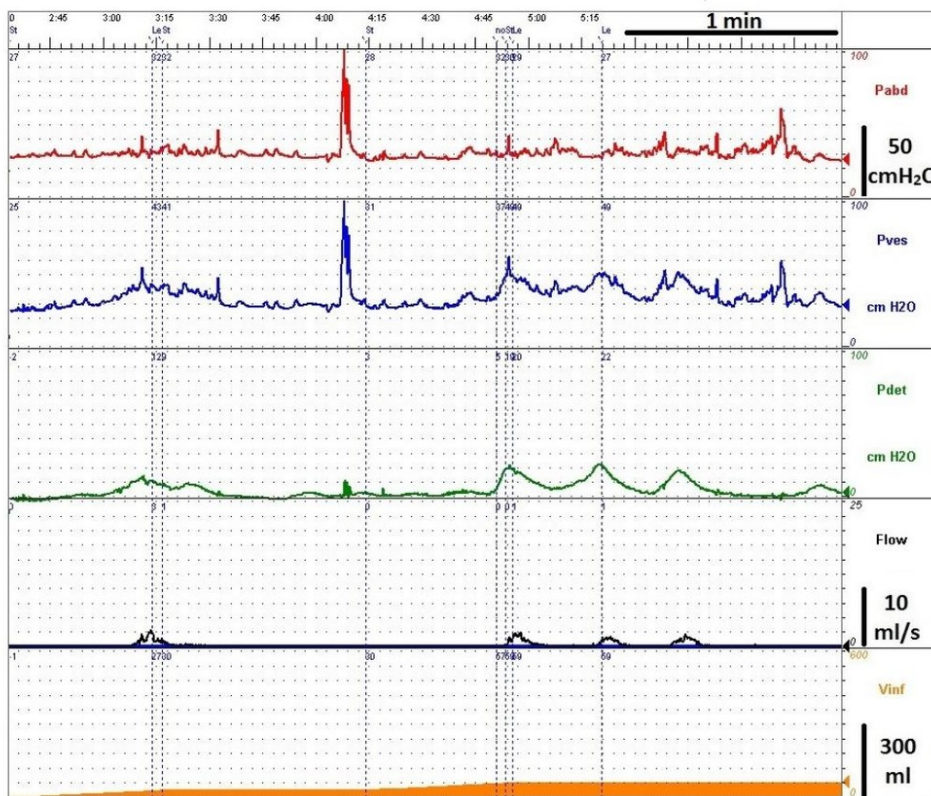


Figure L1.15 — Urodynamics in storage phase: stress UI (3 coughs → 2 leak spikes)

gynecology, obstetrics, and neonatology



Detrusor pressure
(overactive)

Flow
(incontinence)

Urodynamics test in the storage phase, showing 4 episodes of *detrusor overactivity* (detrusor should be inactive) with *incontinence* (there should be no urine flow during storage)

Figure L1.16 — Urodynamics: detrusor overactivity (Pdet rises) with incontinence in storage

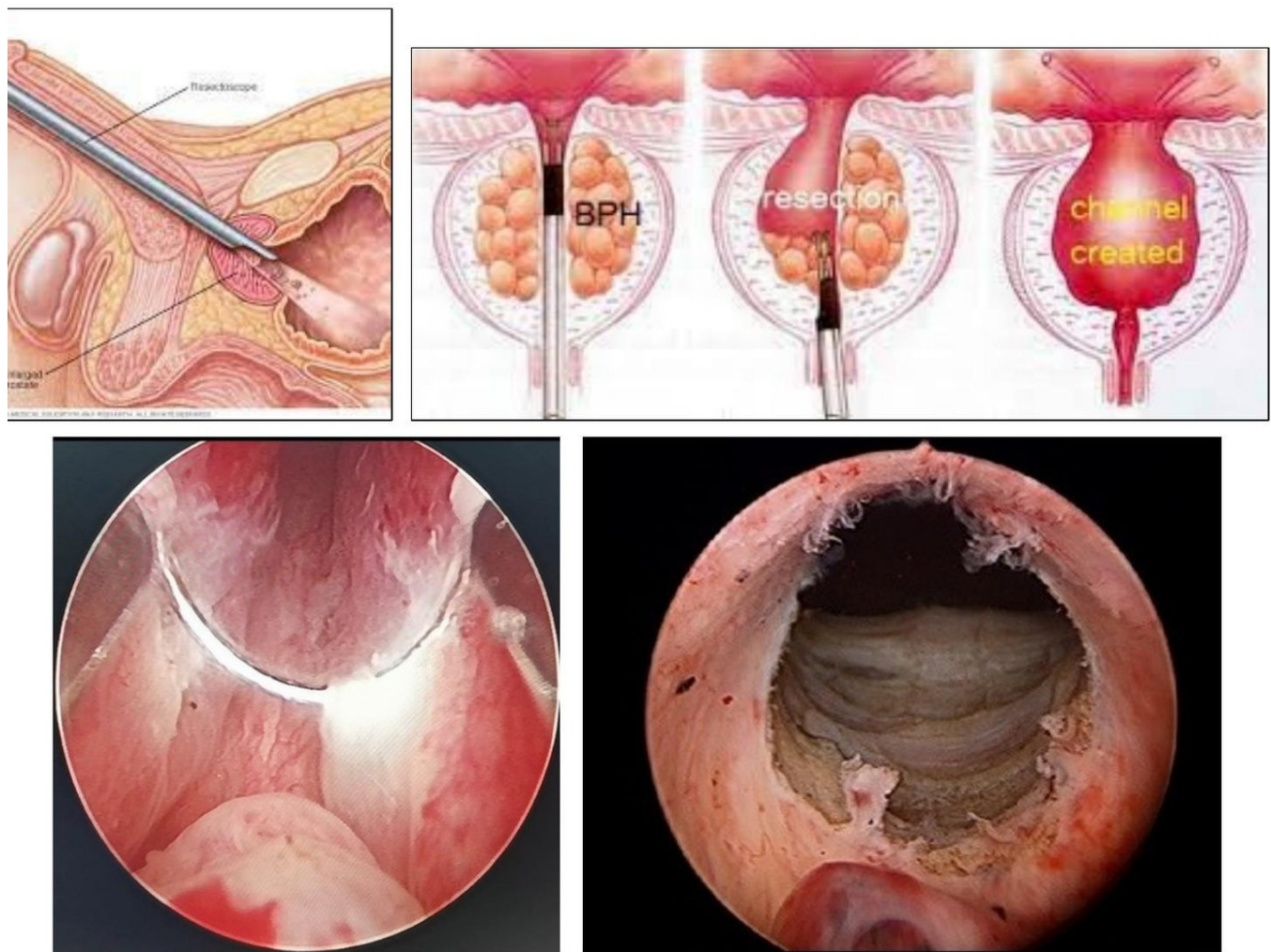


Figure L1.17 — TURP: endoscopic resection of prostate, before/after cystoscopic appearance (lateral lobes)